Weakly Compact Multipliers Between Abstract Tandori Function Spaces

Candidate full solution packet for Problem 7.5

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Abstract

Let $I = (0,1)$ or $I = (0,\infty)$ with Lebesgue measure, and let X, Y be Köthe function spaces on I. For the associated abstract Tandori spaces $\widetilde{X}, \widetilde{Y}$, we prove that every bounded multiplication operator $M_\lambda : \widetilde{X} \to \widetilde{Y}$ is maximally far from both the weakly compact and the compact operators:

$$\operatorname{dist}\big(M_\lambda, \mathcal{W}(\widetilde{X}, \widetilde{Y})\big) = \operatorname{dist}\big(M_\lambda, \mathcal{K}(\widetilde{X}, \widetilde{Y})\big) = \|M_\lambda\|.$$

Consequently, M_λ is weakly compact, or compact, if and only if $\lambda = 0$ almost everywhere. In fact, every nonzero multiplier fixes an isometric copy of ℓ_∞ and hence is not strictly singular. The key ingredient for the exact norm formula is a simultaneous tail-preserving disjointization lemma: any two nonnegative measurable functions can be cut into countably many pairwise disjoint pieces, each of which has exactly the same least decreasing majorant as the original function.

1 Packet status and source evidence

This packet upgrades the earlier compact-only partial packet for the same source problem. That packet is preserved in the local history directory, and the supplied TeX/PDF report is stored in a dated evidence directory.

The source target is Problem 7.5 of Kiwerski–Tomaszewski:

of symbols that generate strictly singular and nuclear multipliers.

7.6. **Compact multipliers between Tandori function spaces.** For a given Köthe function space X, by the **abstract Tandori function space** $\widetilde{X}$ we mean a vector space of all real-valued measurable functions, say f, with $\widetilde{f} \in X$, where $\widetilde{f}$ is the **least decreasing majorant** of a given function f, that is, $\widetilde{f}(x) := \operatorname{ess\,sup}_{t \geqslant x} |f(t)|$. We equip $\widetilde{X}$ with the norm $\|f\|_{\widetilde{X}} := \|\widetilde{f}\|_X$. It is straightforward to see that the ideal $(\widetilde{X})_o$ is trivial (see, for example, [LM15, Theorem 1(e)]; in fact, due to [KKM21, Proposition 4.8], the space $\widetilde{X}$ even contains a lattice isometric copy of ℓ_∞). Thus, in light of Theorem 4.1 and Theorem 4.16, we want to pose the following

Problem 7.5. *What can we say about (weakly) compact multipliers between $\widetilde{X}$ and $\widetilde{Y}$?*

2 The problem and the main result

Throughout, $I = (0,1)$ or $I = (0,\infty)$, equipped with Lebesgue measure m. For a measurable function f on I, put

$$\widetilde{f}(x) := \operatorname*{ess\,sup}_{t > x} |f(t)|, \qquad x \in I.$$

(The harmless alternative $t \geq x$ gives the same function because singletons are null.) If X is a Köthe function space on I, its abstract Tandori space is

$$\widetilde{X} := \{f : \widetilde{f} \in X\}, \qquad \|f\|_{\widetilde{X}} := \left\|\widetilde{f}\right\|_X .$$

We use the convention of Kiwerski–Tomaszewski that $\mathbf{1}_A \in X$ whenever $m(A) < \infty$, and similarly for Y.

For Banach spaces E, F, let $\mathcal{K}(E,F)$ and $\mathcal{W}(E,F)$ denote, respectively, the compact and weakly compact operators from E to F. The essential and weak essential norms of $T \in \mathcal{L}(E,F)$ are

$$\|T\|_{\mathrm{ess}} := \mathrm{dist}(T, \mathcal{K}(E,F)), \qquad \|T\|_{\mathrm{wess}} := \mathrm{dist}(T, \mathcal{W}(E,F)).$$

The following gives a complete answer to Problem 7.5 in the source paper [1, Problem 7.5].

Theorem 2.1 (Exact weak and compact essential norms)**.** *Let X and Y be Köthe function spaces on I, and suppose that $M_\lambda : \widetilde{X} \to \widetilde{Y}$ is bounded. Then*

$$\|M_\lambda\|_{\mathrm{wess}} = \|M_\lambda\|_{\mathrm{ess}} = \|M_\lambda\| . \tag{1}$$

More precisely, for every nonzero $f \in \widetilde{X}$ there is a linear isometry $J_f : \ell_\infty \to \widetilde{X}$ such that

$$\|M_\lambda J_f a\|_{\widetilde{Y}} = \frac{\|M_\lambda f\|_{\widetilde{Y}}}{\|f\|_{\widetilde{X}}} \|a\|_{\ell_\infty} \qquad (a \in \ell_\infty). \tag{2}$$

Corollary 2.2. *For a bounded multiplier $M_\lambda : \widetilde{X} \to \widetilde{Y}$, the following are equivalent:*
- (i) *M_λ is weakly compact;*
- (ii) *M_λ is compact;*
- (iii) *$M_\lambda = 0$;*
- (iv) *$\lambda = 0$ almost everywhere.*

Every nonzero M_λ fixes an isometric copy of ℓ_∞ and is therefore not strictly singular.

No Fatou property, rearrangement invariance, separability, reflexivity, or order continuity is required.

3 Proof intuition

The Tandori norm remembers, for each point x, the largest value appearing to the right of x. This makes it possible to split a function into many disjoint pieces without changing its least decreasing majorant, as long as each piece still meets every right-tail level set. The countable disjointization lemma performs this split simultaneously for $|f|$ and $|\lambda f|$.

Once such pieces are available, the map $a = (a_n) \mapsto \sum_n a_n f \mathbf{1}_{A_n}$ embeds ℓ_∞ isometrically into $\widetilde{X}$, and multiplication by λ acts on that copy with exactly the scale $\|M_\lambda f\|_{\widetilde{Y}} / \|f\|_{\widetilde{X}}$. Any weakly compact operator within a smaller distance from M_λ would be bounded below on an ℓ_∞ copy, impossible because bounded-below weakly compact operators have reflexive domain. Taking near-extremizing f gives the exact weak essential norm, and compact operators are a subclass of weakly compact operators.

4 Assessment and completion of the supplied partial argument

The supplied compactness obstruction is correct. Its only informal step is the construction of disjoint subsets having a common essential right endpoint; the following elementary lemma makes that step precise.

Lemma 4.1 (Common-endpoint partition)**.** *Let $E \subset I$ have positive measure and finite essential supremum $b := \operatorname{ess\,sup} E$. Then E contains pairwise disjoint measurable sets $B_1, B_2, \ldots$ such that $\operatorname{ess\,sup} B_n = b$ for every n.*

Proof. Choose $r_1 \in (0, b)$. Having chosen $r_k < b$, the set $E \cap (r_k, b)$ has positive measure and is the increasing union of the sets $E \cap (r_k, s)$ with rational $s < b$. Hence some $s_k < b$ satisfies $m(E \cap (r_k, s_k)) > 0$. Choose

$$\max\{r_k, s_k, b - 1/(k+1)\} < r_{k+1} < b.$$

Then $r_k \uparrow b$ and $m(E \cap (r_k, r_{k+1})) > 0$ for every k. Partition $\mathbb{N}$ into pairwise disjoint infinite sets $N_1, N_2, \ldots$, and set

$$B_n := \bigcup_{k \in N_n} E \cap (r_k, r_{k+1}).$$

Each N_n is unbounded, so each B_n has essential supremum b. □

The partial proof then normalizes the functions $\mathbf{1}_{B_n}$ and obtains a bounded sequence whose images are uniformly separated. That proves noncompactness. The same construction actually gives more: it produces an entire copy of ℓ_∞, which also rules out weak compactness.

Lemma 4.2. *Let $S : E \to F$ be a weakly compact operator between Banach spaces. If S is bounded below, then E is reflexive. In particular, no weakly compact operator defined on ℓ_∞ is bounded below.*

Proof. If $\|Sx\| \geq c\,\|x\|$ for some $c > 0$, then $S(E)$ is closed and $S : E \to S(E)$ is an isomorphism. Moreover, $S(B_E)$ is norm closed and convex, hence weakly closed. Since S is weakly compact, $S(B_E)$ is therefore weakly compact. The bounded inverse $S^{-1} : S(E) \to E$ is weak-to-weak continuous (the weak topology of a closed subspace is the induced weak topology, by Hahn–Banach), so $B_E = S^{-1}(S(B_E))$ is weakly compact. Thus E is reflexive. Since ℓ_∞ is not reflexive, the final claim follows. □

Proposition 4.3 (The partial argument already yields the qualitative answer)**.** *If $M_\lambda : \widetilde{X} \to \widetilde{Y}$ is nonzero, then it is bounded below on an isometric copy of ℓ_∞. Hence it is neither weakly compact nor compact.*

Proof. Since $M_\lambda \neq 0$, the symbol is nonzero on a set of positive measure. Choose $\varepsilon > 0$ and $N < \infty$ so that

$$E := \{t \in I : |\lambda(t)| > \varepsilon\} \cap (0, N)$$

has positive measure. Put $b = \operatorname{ess\,sup} E$, and use Lemma 4.1 to obtain disjoint $B_n \subset E$ with $\operatorname{ess\,sup} B_n = b$. Let $c_X = \left\|\mathbf{1}_{(0,b)}\right\|_X$ and define

$$Ja := \frac{1}{c_X} \sum_{n=1}^{\infty} a_n \mathbf{1}_{B_n}, \qquad a = (a_n) \in \ell_\infty.$$

The sum is pointwise unambiguous because the B_n are disjoint. For every $a \in \ell_\infty$,

$$\widetilde{Ja} = \frac{\|a\|_\infty}{c_X}\mathbf{1}_{(0,b)},$$

so J is an isometry. Moreover,

$$\widetilde{\lambda Ja} \geq \frac{\varepsilon\,\|a\|_\infty}{c_X}\mathbf{1}_{(0,b)}.$$

Consequently,

$$\|M_\lambda Ja\|_{\widetilde{Y}} \geq \varepsilon\frac{\left\|\mathbf{1}_{(0,b)}\right\|_Y}{\left\|\mathbf{1}_{(0,b)}\right\|_X}\|a\|_\infty.$$

Thus $M_\lambda J$ is bounded below. If M_λ were weakly compact, then so would be $M_\lambda J$, contradicting Lemma 4.2. ☐

This proves Corollary 2.2 qualitatively. To obtain the sharp identities in Theorem 2.1, the copy of ℓ_∞ must be adapted not only to a level set of λ, but simultaneously to an arbitrary near-extremizer f and its image λf.

5 Simultaneous tail-preserving disjointization

We first prove a measure-theoretic splitting lemma. A finite measure ν is called equivalent to Lebesgue measure if it has the same null sets. On $(0,\infty)$, for example, one may take $d\nu(t) = e^{-t}\,dt$.

Lemma 5.1 (Countable simultaneous splitting)**.** *Let (Ω, Σ, ν) be an atomless finite measure space, and let $E_1, E_2, \ldots \in \Sigma$ satisfy $\nu(E_k) > 0$ for every k. There are pairwise disjoint measurable sets $A_1, A_2, \ldots$ such that*

$$\nu(A_n \cap E_k) > 0 \qquad (n, k \in \mathbb{N}).$$

Proof. We first show that any measurable set P satisfying $\nu(P \cap E_k) > 0$ for all k can be divided into two disjoint pieces, each of which still meets every E_k in positive measure.

Work in the measure algebra of P, equipped with the complete metric $d(A,B) = \nu(A \triangle B)$. For each k, put $C_k = P \cap E_k$ and define

$$\mathcal{U}_k := \{A \subset P : 0 < \nu(A \cap C_k) < \nu(C_k)\}.$$

The set $\mathcal{U}_k$ is open. It is also dense: if a given $A \subset P$ does not already split C_k, then either $A \cap C_k$ or $C_k \setminus A$ is null. Atomlessness allows us to add or remove an arbitrarily small positive-measure subset of C_k, thereby producing a nearby set that does split C_k. By the Baire category theorem, $\bigcap_{k=1}^\infty \mathcal{U}_k$ is nonempty. Any A in this intersection gives the desired division $P = A \sqcup (P \setminus A)$.

Starting with $P_\varnothing = \Omega$, apply this division recursively to construct a binary tree $(P_s)_{s\in\{0,1\}^{<\mathbb{N}}}$ such that $P_s = P_{s0} \sqcup P_{s1}$ and every P_s meets every E_k in positive measure. Finally, set

$$A_n := P_{\underbrace{11\ldots1}_{n-1}0} \qquad (n \geq 1).$$

These sets are pairwise disjoint, and each one meets every E_k in positive measure. ☐

Lemma 5.2 (Tail-preserving disjointization)**.** *Let $g, h : I \to [0,\infty]$ be measurable. There are pairwise disjoint measurable sets $A_1, A_2, \ldots \subset I$ such that, for every n,*

$$\widetilde{g\mathbf{1}_{A_n}} = \widetilde{g}, \qquad \widetilde{h\mathbf{1}_{A_n}} = \widetilde{h} \quad \textit{almost everywhere on } I. \tag{3}$$

Proof. Choose an atomless finite measure ν equivalent to Lebesgue measure. Form the countable family consisting of all positive-measure sets of the form

$$E^g_{r,q} := \{t > r : g(t) > q\}, \qquad E^h_{r,q} := \{t > r : h(t) > q\},$$

where $r \in \mathbb{Q} \cap I$ and $q \in \mathbb{Q}_{>0}$; discard the null sets. Apply Lemma 5.1 to this family, repeating its members if it is finite. (If the family is empty, take $A_n = \varnothing$.) We obtain disjoint A_n meeting every retained set in positive measure.

Fix n and $x \in I$. The inequality $\widetilde{g\mathbf{1}_{A_n}}(x) \le \widetilde{g}(x)$ is immediate. Conversely, let $0 \le \alpha < \widetilde{g}(x)$. Choose rational q with $\alpha < q < \widetilde{g}(x)$. Then $\{t > x : g(t) > q\}$ has positive measure. Since

$$\{t > x : g(t) > q\} = \bigcup_{\substack{r \in \mathbb{Q}\cap I \\ r > x}} \{t > r : g(t) > q\},$$

some retained $E^g_{r,q}$ has positive measure. Its intersection with A_n has positive measure, and therefore $\widetilde{g\mathbf{1}_{A_n}}(x) \ge q > \alpha$. Letting α increase to $\widetilde{g}(x)$ gives equality. The argument for h is identical. $\square$

Proposition 5.3 (An adapted isometric copy of ℓ_∞)**.** *Let $f \in \widetilde{X}$ be nonzero and let $M_\lambda : \widetilde{X} \to \widetilde{Y}$ be bounded. There are pairwise disjoint measurable sets A_n such that the map*

$$J_f(a) := \frac{1}{\|f\|_{\widetilde{X}}} \sum_{n=1}^{\infty} a_n f \mathbf{1}_{A_n} \qquad (a = (a_n) \in \ell_\infty) \tag{4}$$

is a linear isometry from ℓ_∞ into $\widetilde{X}$ and satisfies (2).

Proof. Apply Lemma 5.2 to $g = |f|$ and $h = |\lambda f|$. Put $f_n = f\mathbf{1}_{A_n}$. Then

$$\widetilde{f_n} = \widetilde{f}, \qquad \widetilde{\lambda f_n} = \widetilde{\lambda f} \quad (n \ge 1).$$

For $a \in \ell_\infty$, disjointness gives the pointwise estimates

$$\left|\sum_n a_n f_n\right| \le \|a\|_\infty |f|, \qquad \left|\lambda \sum_n a_n f_n\right| \le \|a\|_\infty |\lambda f|.$$

On the other hand, for every fixed n the two sums dominate, on A_n, $|a_n||f|$ and $|a_n||\lambda f|$, respectively. Hence their least decreasing majorants dominate $|a_n|\widetilde{f}$ and $|a_n|\widetilde{\lambda f}$ for every n. Taking the supremum over n and combining with the preceding upper estimates yields the exact identities

$$\widetilde{\sum_n a_n f_n} = \|a\|_\infty \widetilde{f}, \qquad \widetilde{\lambda \sum_n a_n f_n} = \|a\|_\infty \widetilde{\lambda f}.$$

After the normalization in (4), these are precisely the asserted isometry and formula (2). $\square$

6 Proof of the exact norm formula

Proof of Theorem 2.1. Let $T = M_\lambda$ and fix $0 \ne f \in \widetilde{X}$. Take the isometry $J_f : \ell_\infty \to \widetilde{X}$ from Proposition 5.3, and set

$$c_f := \frac{\|Tf\|_{\widetilde{Y}}}{\|f\|_{\widetilde{X}}}.$$

Then $\|TJ_f a\| = c_f \|a\|_\infty$ for all $a \in \ell_\infty$.

Let $W \in \mathcal{W}(\widetilde{X}, \widetilde{Y})$. If $\|T - W\| < c_f$, then

$$\|WJ_f a\| \geq \|TJ_f a\| - \|(T - W)J_f a\| \geq (c_f - \|T - W\|) \|a\|_\infty .$$

Thus $WJ_f : \ell_\infty \to \widetilde{Y}$ would be weakly compact and bounded below, contrary to Lemma 4.2. Hence

$$\|T - W\| \geq c_f \qquad \text{for every } 0 \neq f \in \widetilde{X}.$$

Taking the supremum over f gives

$$\|T - W\| \geq \|T\| .$$

Since the zero operator is weakly compact, the reverse inequality for the distance is automatic. Therefore

$$\operatorname{dist}(T, \mathcal{W}(\widetilde{X}, \widetilde{Y})) = \|T\| .$$

Finally, $\mathcal{K}(\widetilde{X}, \widetilde{Y}) \subset \mathcal{W}(\widetilde{X}, \widetilde{Y})$, so

$$\|T\| = \operatorname{dist}(T, \mathcal{W}) \leq \operatorname{dist}(T, \mathcal{K}) \leq \|T\| ,$$

which proves (1). □

Proof of Corollary 2.2. The equivalence of weak compactness, compactness, and $M_\lambda = 0$ follows immediately from Theorem 2.1. If $\lambda \neq 0$ on a set of positive measure, choose $b < \infty$ so that $\lambda \mathbf{1}_{(0,b)} \neq 0$. Since $\mathbf{1}_{(0,b)} \in \widetilde{X}$, this shows $M_\lambda \neq 0$. Thus $M_\lambda = 0$ if and only if $\lambda = 0$ almost everywhere.

If $M_\lambda \neq 0$, choose f with $M_\lambda f \neq 0$. Formula (2) shows that the restriction of M_λ to $J_f(\ell_\infty)$ is bounded below, hence an isomorphism onto its range. Therefore M_λ is not strictly singular. □

The same argument gives a small additional strengthening.

Corollary 6.1 (Distance from strictly singular operators)**.** *For every bounded multiplier $M_\lambda : \widetilde{X} \to \widetilde{Y}$,*

$$\operatorname{dist}\big(M_\lambda, \mathcal{SS}(\widetilde{X}, \widetilde{Y})\big) = \|M_\lambda\| ,$$

where $\mathcal{SS}$ denotes the ideal of strictly singular operators.

Proof. If S is strictly singular and $\|M_\lambda - S\| < c_f$, the same estimate used above makes SJ_f bounded below on ℓ_∞, contradicting strict singularity. Take the supremum over f and use $0 \in \mathcal{SS}$. □

7 Literature check and relation to earlier structure results

The accessible source version is arXiv:2212.06723v5, last revised in March 2024; it explicitly poses the compact/weakly compact Tandori multiplier question as Problem 7.5 [1]. The paper was published online in *Mathematische Nachrichten* in June 2026. Searches performed through 21 June 2026 for the exact problem number, the phrases "compact multipliers between Tandori function spaces" and "weakly compact multipliers Tandori", and papers citing the source did not locate a published or preprint solution.

A relevant earlier structural theorem of Kiwerski–Kolwicz–Maligranda says that every nontrivial abstract Tandori function space contains an order isometric copy of ℓ_∞ [2, Proposition 4.8]. That fact alone does not force a particular multiplier to be bounded below on that copy. The common-endpoint construction in Proposition 4.3 localizes the copy inside a level set of the symbol, while Lemma 5.2

adapts it simultaneously to f and λf; this extra localization is what produces the exact weak essential norm.

The later work on arithmetic and factorization of spaces related to decreasing functions develops formulas for bounded multiplier spaces and factorizations involving Tandori spaces [3], but does not address this weak compactness obstruction. Thus the theorem above appears to supply the missing full answer to the stated problem; this novelty claim is necessarily qualified by the limits of the literature search.

8 Human review recommendation

Review as a candidate full solution. The main points to check are the tail-preserving disjointization lemma, the exact least-decreasing-majorant identities in Proposition 5.3, and whether the source paper's conventions for Köthe function spaces cover every indicator and non-atomic splitting step used here. The literature status should also be checked against later multiplier papers and the 2026 published version.

9 Conclusion

The partial compactness proof is valid, but it stops one step before the strongest consequence of its own construction. Its disjoint common-endpoint sets span an isometric ℓ_∞ on which every nonzero multiplier is bounded below, already excluding weak compactness. A simultaneous version of the same idea then yields the quantitative identity

$$\boxed{\|M_\lambda\|_{\mathrm{wess}} = \|M_\lambda\|_{\mathrm{ess}} = \|M_\lambda\|\,.}$$

References

[1] T. Kiwerski and J. Tomaszewski, *Essential norms of pointwise multipliers in the non-algebraic setting*, Mathematische Nachrichten (2026), 1–39, DOI: 10.1002/mana.70174; accessible preprint: arXiv:2212.06723v5.

[2] T. Kiwerski, P. Kolwicz and L. Maligranda, *Isomorphic and isometric structure of the optimal domains for Hardy-type operators*, Studia Math. **260** (2021), 45–89; arXiv:1906.09672.

[3] T. Kiwerski and J. Tomaszewski, *Arithmetic, interpolation and factorization of amalgams*, arXiv:2401.05526v2 (2024).

[4] K. Leśnik, L. Maligranda and J. Tomaszewski, *Weakly compact sets and weakly compact pointwise multipliers in Banach function lattices*, Math. Nachr. **295** (2022), 574–592.